

Patrick Hallaert

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EDUCATION

Emory University School of Medicine
Doctor of Medicine (MD)

2025-2029
Atlanta, GA, USA

McGill University
Bachelor of Science (BS), Honors Biology

2019-2023
Montreal, QC, Canada

RESEARCH EXPERIENCE

Emory University School of Medicine
Medical Student Researcher in Dr. Hwa Yeung's Lab

December 2025 – Present
Atlanta, GA

- Leading a qualitative study examining resident research training in dermatology, using transcript coding and thematic analysis to characterize residents' experiences with research mentorship, engagement, and scholarly development.
- Contributing to ongoing research projects while strengthening skills in chart review, systematic review development, literature review, and manuscript revision.

National Institutes of Health
Postbaccalaureate Research Fellow in Dr. Isaac Brownell's Lab

May 2023 – June 2025
Bethesda, MD

- Established a novel mouse model for virus-positive Merkel cell carcinoma while simultaneously managing my lab's mouse colony, which comprised hundreds of mice across over 50 distinct genetic strains.
- Contributed to multiple other research projects, ranging from an investigation of the immunological mechanisms underlying spontaneous regression in Merkel cell carcinoma (MCC) to a study analyzing how racial and ethnic differences influence survival outcomes in patients with MCC.
- Gained firsthand exposure to clinical dermatology by shadowing physicians in the dermatology clinic, frequenting dermatology grand rounds, and attending cutaneous oncology tumor boards.

McGill University
Honors Researcher in Dr. David Dankort's Lab

September 2022 – April 2023
Montreal, QC

- Aimed to uncover ablated kinases that resensitize drug-resistant lung adenocarcinoma (LUAC) cells to trametinib, a MEK inhibitor drug.
- Conducted a kinase-targeting CRISPR screen on mouse-derived, trametinib-resistant LUAC cells with p53 loss and oncogenic Kras or Braf mutations.

Harvard University
Research Assistant in Dr. Quan Lu's Lab

July 2022
Boston, Massachusetts

- Prepared ARRDC1-mediated microvesicles (ARMMs) by transfecting HEK293 cells with a plasmid containing a shortened ARRDC1-Cas9 fusion protein targeting the GFP gene.
- After collecting and quantifying the microvesicles, recipient U2OS-GFP cells were treated with the ARMMs.
- Genome editing efficacy was determined via flow cytometry and sequencing; the use of these ARMMs resulted in a decrease in GFP-positive cells, demonstrating that gene editing likely occurred.

Harvard University

May - June 2022

- Assisted with the development of FAVOR (Functional Annotation of Variants Online Resource), which gathers, outlines, and visualizes functional annotation data.
- Employed Python, R, and the command line to create a back-end functional annotation database for the GRCh37 / hg19 reference genome.
- Optimized database code to prioritize efficient memory consumption and low computing time; this process included correcting memory leaks and implementing techniques such as lazy loading and sparse data structures.

PUBLICATIONS

Hallaert P, Roman JW, Baker M, Hill NT, Marte JL, Gulley JL, Logemann N, Hudgens CW, Reuben A, Brownell I. Spontaneous tumor regression and immunotherapy response demonstrate clonal T-cell expansion in Merkel cell carcinoma. *NPJ Precis Oncol.* 2025 Jul 1;9(1):212. doi: 10.1038/s41698-025-00987-6. PMID: 40594889; PMCID: PMC12218097.

Chefitz G, Martin MR, Vilasi SM, Saito Y, Jarvis JE, **Hallaert P**, Reed DJ, Shalhout SZ, Brownell I. Merkel Cell Carcinoma Virus Positivity Ranges From 57% to 77% in the United States Based on Geographic Region: A Pooled Cohort Study. *J Am Acad Dermatol.* 2026 May 28:S0190-9622(26)02779-9. doi: 10.1016/j.jaad.2026.05.081. Epub ahead of print. PMID: 42214499.

Miao L, Collado L, Barkdull S, **Hallaert P**, Martin MR, Gryder BE, Kelly MC, Dell'Orso S, Kelley MW, Brownell I. Single cell analysis of developing Merkel cells reveals the emergence of non-coding RNA biotypes as a hallmark of terminal differentiation. *Cell Death Differ.* 2026 Feb 3. doi: 10.1038/s41418-026-01663-3. Epub ahead of print. PMID: 41634368.

Martin MR, Vilasi SM, Saito Y, Mohsin N, Jarvis JE, **Hallaert P**, Reed DJ, Miao L, Tribble JT, Cahoon EK, Pfeiffer RM, Lachance K, Kulikauskas R, Nghiem P, Brownell I. Population differences in Merkel cell carcinoma by virus status and anatomic site: A multi-cohort analysis including institutional, SEER, and NCDB data. *J Am Acad Dermatol.* 2025 Nov 14:S0190-9622(25)03221-9. doi: 10.1016/j.jaad.2025.11.023. Epub ahead of print. PMID: 41242630.

Strong J, Winkie MJ, **Hallaert P**, Whitecar SB, Keung ES, Neelon D, Zeman KG, Gage MM, Schaffenburg WC, Simpson MM, Brownell I. Epidermotropic Metastatic Melanoma Presenting as Eruptive Primary Melanomas. *Pigment Cell Melanoma Res.* 2025 Jul;38(4):e70037. doi: 10.1111/pcmr.70037. Epub 2025 Jul 4. PMCID: PMC12265876.

Strong J, Adhanom R, Kim CS, Saito Y, Meltzer JC, **Hallaert P**, Martinez S, Salancy A, Kong HH, Cowen EW, Castelo-Soccio L, Murphy PM, McDermott DH, Brownell I. Risk of Superficial Fungal Infections in WHIM Syndrome. *Dermatol Ther (Heidelb).* 2025 May;15(5):1173-1179. doi: 10.1007/s13555-025-01396-0. Epub 2025 Apr 3. PMID: 40178760; PMCID: PMC12033133.

Strong J, Poon VI, **Hallaert P**, Francis D, Pastor DM, Castelo-Soccio L, Strauss J, Brownell I. Alopecia and hair repigmentation associated with anti-programmed death-ligand 1 (PD-L1) immunotherapy. *JAAD Case Rep.* 2024 Nov 19;56:51-53. doi: 10.1016/j.jidcr.2024.09.029. PMID: 39839462; PMCID: PMC11750434.

Jarvis JE, Miao L, **Hallaert P**, Martin MR, Collado L, Hill NT, Brownell I. POU4F3 Is Necessary for Normal Merkel Cell Formation, whereas POU4F1 Is Dispensable. *J Invest Dermatol.* 2025 Jan;145(1):193-196.e5. doi: 10.1016/j.jid.2024.06.1284. Epub 2024 Jul 14. PMID: 39009281; PMCID: PMC12327023.

Strong J, **Hallaert P**, Brownell I. Merkel Cell Carcinoma. *Hematol Oncol Clin North Am.* 2024 Oct;38(5):1133-1147. doi: 10.1016/j.hoc.2024.05.013. Epub 2024 Jul 26. PMID: 39060119; PMCID:

PMC11423797.

Garman KA, Thoreson N, Strong J, **Hallaert P**, Gelb T, Shen M, Hall MD, Brownell I. Mycophenolate mofetil inhibits Merkel cell carcinoma growth. *Br J Dermatol*. 2024 Mar 15;190(4):593-595. doi: 10.1093/bjd/ljae018. PMID: 38266271; PMCID: PMC10941324.

ABSTRACTS

Jarvis JE, Miao L, **Hallaert P**, Martin MR, Collado L, Hill NT, Brownell I. 408 POU4F3 and POU4F1 are expressed in mouse Merkel cells but only POU4F3 is critical for their development. *Journal of Investigative Dermatology*, Volume 144, Issue 8, Supplement, 2024, Page S71, ISSN 0022-202X, <https://doi.org/10.1016/j.jid.2024.06.424>

POSTER AND ORAL PRESENTATIONS

Hallaert P, Strong J, Winkie MJ, Whitecar SB, Keung ES, Neelon D, Zeman KG, Gage MM, Schaffenburg WC, Simpson MM, Brownell I. "Differentiating epidermotropic metastatic melanoma from primary melanomas through panel sequencing." National Institute of Arthritis and Musculoskeletal and Skin Diseases Scientific Retreat, May 2025. Bethesda, MD, USA.

Hallaert P, Strong J, Winkie MJ, Whitecar SB, Keung ES, Neelon D, Zeman KG, Gage MM, Schaffenburg WC, Simpson MM, Brownell I. "Applying next generation sequencing to distinguish epidermotropic metastatic melanoma from eruptive primary melanomas." Poster presented at the National Institutes of Health Postbac Poster Day, May 2025. Bethesda, MD, USA.

Hallaert P, Baker M, Roman JW, Reuben A, Hill N, Hudgens CW, Tetzlaff MT, Logemann NF, and Brownell I. "Clonal T cell expansion drives spontaneous tumor regression in Merkel cell carcinoma." Poster presented at the National Institutes of Health Research Festival, September 2024. Bethesda, MD, USA.

Hallaert P, Baker M, Roman JW, Reuben A, Hill N, Hudgens CW, Tetzlaff MT, Logemann NF, and Brownell I. "Spontaneous regression of Merkel cell carcinoma is driven by clonal T cell expansion." Poster presented at the National Institute of Arthritis and Musculoskeletal and Skin Diseases Intramural Research Program Scientific Retreat, June 2024. Bethesda, MD, USA.

- **Awarded the 2024 Best Poster Presentation Award.**

Hallaert P, Baker M, Roman JW, Reuben A, Hill N, Hudgens CW, Tetzlaff MT, Logemann NF, and Brownell I. "A Vanishing Act: How Your Body Can Naturally Kill Cancer." Oral presentation presented at the NIH National Institute of Arthritis and Musculoskeletal and Skin Diseases Three-Minute Talk Competition, May 2024. Bethesda, MD, USA.

Hallaert P, Baker M, Roman JW, Reuben A, Hill N, Hudgens CW, Tetzlaff MT, Logemann NF, and Brownell I. "Adaptive T cell expansion drives spontaneous regression in Merkel cell carcinoma." Poster presented at the National Institutes of Health Postbac Poster Day, May 2024. Bethesda, MD, USA.

- **Awarded the 2024 Outstanding Poster Award.**

Hallaert P, Miao LL, Martin M, Collado L, Barkdull S, Coxon A, Gryder B, Brownell I. "Gfi1 controls Merkel cell survival through a dependence receptor pathway." Poster presented at the National Institutes of Health Research Festival, September 2023. Bethesda, MD, USA.

Hallaert P, Miao LL, Martin M, Collado L, Barkdull S, Coxon A, Gryder B, Brownell I. "Gfi1 regulates dependence receptors to prevent suicide in sensory Merkel cells." Poster presented at the National Institute of Arthritis and Musculoskeletal and Skin Diseases Scientific Retreat, June 2023. Bethesda, MD, USA.

Hallaert P, Dankort D. “Exploring Trametinib Resistance in Lung Adenocarcinoma: Which Ablated Kinases Could Function Synergistically with MEK1/2 Inhibition to Prevent Cancer Recurrence?” Oral presentation presented at the McGill University Honors Biology Symposium, April 2023. Montreal, QC, Canada.

LEADERSHIP

Treasurer **May 2026 - Present**
Cutaneous Oncology Interest Group, Emory University School of Medicine *Atlanta, GA*

- Currently helping establish a novel, student-led interest group focused on cutaneous oncology.
- Helping develop plans for future educational programming, including “day in the life” sessions, career panels, and clinical skill workshops.

Advocacy Co-Chair **Oct 2025 - Present**
Asian Pacific American Medical Student Association *Atlanta, GA*

- Coordinating APAMSA's service activities, including helping organize a medical school-wide bone marrow donation event in collaboration with the National Marrow Donor Program (NMDP).
- Establishing connections with Atlanta-based nonprofits to foster collaboration and connect medical students with meaningful community engagement opportunities.

NIH Postbaccalaureate Committee **May 2024 - June 2025**
Committee Co-Chair *Bethesda, MD*

- Elected as the co-chair of the entire Postbaccalaureate Committee after previously demonstrating leadership and dedication as the co-chair of the social subcommittee.
- As the co-chair, I oversaw 6 subcommittees and 3 liaisons that catered to the diverse needs of 1000+ research fellows.
- Some of the committee's roles include running a seminar series, offering health and wellness resources, facilitating community service opportunities, and organizing events.

Social Subcommittee Co-Chair **October 2023 – May 2024**

- As the social subcommittee co-chair, I was responsible for organizing academic, social, and networking events for postbaccalaureate fellows.

Organizing Health Equity Symposium **October 2023 – May 2024**
In collaboration with NIH and FDA representatives *Bethesda, MD*

- Organized a health equity symposium focusing on initiatives that aim to increase diversity in clinical research; event featured keynote speaker from the FDA that is an expert in the field.
- Collaborated with the *National Institutes of Health Academy on Health Disparities*, which has 300+ yearly members.

McGill Science Undergraduate Research Journal **October 2022 - April 2023**
Scientific Editor *Montreal, QC*

- Helped coordinate the peer-review process for undergraduate research.
- Edited and assessed submitted manuscripts, working with authors and peer reviewers to improve papers.

Science Undergraduate Society of McGill University **October 2022 - April 2023**
Academic Events Committee Coordinator *Montreal, QC*

- Organized academic events such as symposiums, panels, and networking events for all undergraduate students in the Faculty of Science.
- Helped develop partnerships with over twenty other student organizations.

CLINICAL EXPERIENCE

Geriatric Primary Care Experience

November 2025 - Present

JenCare Senior Medical Center

Atlanta, GA

- Assisting primary care physician in providing comprehensive outpatient care to a diverse, predominantly low-income senior population.
- Conducted focused history-taking, performed physical examinations, and presented assessments to the supervising physician.
- Gained experience serving medically vulnerable older adults with complex comorbidities, emphasizing empathy, access-focused care, and longitudinal patient relationships.

Hospital Volunteer

September 2023 - November 2024

Suburban Hospital

Bethesda, MD

- I assisted clinicians, physician assistants, and nurses in the oncology and medical care units.

Hospital Volunteer

June 2018 - May 2023

Montreal General Hospital and Tourcoing Hospital Center

Montreal, QC & Tourcoing, France

- I assisted medical staff and visitors in the Intensive Care Unit at Montreal General Hospital.
- I assisted patients and healthcare professionals within Tourcoing Hospital Center's cardiology, emergency medicine, and orthopedics department.

COMMUNITY INVOLVEMENT AND SERVICE

Pre-Heath Mentor

Oct 2025 - Present

Emory Pre-Heath Mentorship Program

Atlanta, GA

- Met with pre-medical students at Emory University on a regular basis, providing tailored mentorship and advice.

Tutor

August 2025 - Present

STAR-C After-School Program

Clarkston, GA

- Provided tutoring and academic support to students from diverse immigrant and refugee backgrounds.
- Fostered a culturally-sensitive, welcoming environment that helped students excel as they become acclimated to the American school system.

Crisis Counselor

2020 - 2024

Crisis Text Line

Virtual

- Received 2 weeks of extensive training that taught me crisis management and communication skills.
- As a counselor, I guided, supported, and provided helpful resources to individuals experiencing crisis.
- Effectively communicated with texters managing depression, suicidal thoughts, PTSD, and anxiety.

Tutor

2019 - 2024

Latino Student Fund

Washington, DC

- Tutored low-income and at-risk students in science, math, and English.
- Served as a mentor and helped tutees overcome academic obstacles through tailored teaching methods and positive reinforcement.